



UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

P720/2

TECHNICAL DRAWING

(Mechanical construction)

**SCORING GUIDE
FOR THE SAMPLE PAPER**

ITEM 1:

To attain the highest level of achievement, the candidate is required to:

- Present a focused introduction (With a problem and a solution briefly)
- Present and explain the necessary Personal protective equipment needed in solving the problem in the scenario
- Present all the necessary Materials/tools and equipment needed to solve the problem in the scenario
- Coherently explain the Processes/procedures/Techniques followed in solving the problem in the scenario. Relevant sketches and illustrations are important.
- explain ways of maintaining the product or article from the problem reoccurring.
- Present a focused deduction/conclusion/stand point/finish that relates to the task.

Possible Responses

1. Identified problem and possible solution – Failure of the pedal (Crank system) and the possible solution of fixing a new cotter pin (Explanation of system and how to fix it and maintain it)
2. Personal protective equipment includes:
 - Safety goggles / safety glasses – protect the eyes from flying metal fragments or particles when tapping or removing the cotter pin.
 - Protective gloves – protect hands from cuts, sharp edges, and pinching injuries.
 - Safety boots (steel-toe boots) – protect feet from dropped tools or machine parts.
 - Protective overalls/work clothing – protect the body from dirt, grease, and minor injuries.
 - Hearing protection (earplugs or earmuffs) – if the work is being done in a noisy workshop or around operating machinery.
 - Hard hat/helmet – if working in an area with overhead hazards.
3. Materials.
 - Cotter pin (split pin) – the main fastening material being installed.
 - Replacement nut or bolt (if required) – where the cotter pin secures a nut.
 - Lubricant/grease – helps with smooth fitting and reduces rust.
 - Cleaning cloth/rag – used to clean dirt, grease, or debris from the part.
 - Cleaning solvent (if needed) – for removing oil or rust from components.

4. Tools and equipment

- Pliers (for bending or removing the cotter pin)
- Hammer (for gentle tapping if necessary)
- Spanner/wrench (for tightening or loosening nuts)
- Punch/drift pin (if removing an old cotter pin)

5. processes/procedure followed to fix the cotter pin

- Position the bicycle securely - Place the bicycle on a stand or turn it upside down so it is stable during work.
- Inspect the cotter pin and crank area - Examine the worn or damaged cotter pin and check for any damage around the crank.
- Remove the old cotter pin - Pull it out using pliers or tap it out carefully with a hammer if it is tight.
- Clean the crank and spindle area - Remove dirt, rust, or grease using a cloth and check for wear.
- Insert the new cotter pin - Align the hole properly and insert the new cotter pin in the correct direction.
- Fit and tighten the nut - Replace the nut and tighten it firmly without overtightening.
- Check alignment and security - Ensure the crank is correctly positioned and the cotter pin is properly seated.
- Test the bicycle - Rotate the pedals and ride the bicycle briefly to confirm smooth operation and secure fitting.

6. Maintenance of the window from getting damaged include:

- Regular inspection – check for wear, cracks, bending, or looseness.
- Cleaning and lubrication – keep it free from dirt and apply lubricant to prevent rust and reduce friction.
- Replace damaged cotter pins – change worn or defective cotter pins immediately to maintain safety and proper function.

7. Conclusion

The above measures, when properly followed, will help prevent the identified problems from reoccurring and will improve the durability, safety, and proper use of the bicycle.

NB. All this is presented in an essay form and relevant sketches are very vital.

Item 2

To attain the highest level of achievement, the candidate is required to:

- Correctly laying out the drawing sheet with all required details, including a frame/border line and a title block with clearly printed information.
- Follow the correct procedures to analyse, interpret, and prepare the required information and working drawings using appropriate symbols and conventions.
- Appropriate use of line types, scales, dimensions, and annotations on the drawings presented.
- Produce all required working drawings that can be used in solving a problem that is in the scenario (Fit for purpose)

Possible Responses

1. Correctly Drawing frame and title block printed with:
 - a. Candidate's Identification (name, random and personal no.)
 - b) Title of drawing
 - c) Scales used
 - d) Date
 - e) Client
 - f) Projection symbol
2. Producing required scaled drawings using symbols and conventions, choosing and using the appropriate scale and using suitable line types, annotations and dimensions. The drawings include
 - a) Elevation and plan
 - b) Assembled sectional elevation.
3. Produce the material list.

PART	DESCRIPTION	QUANTITY
1	Base	1
2	Shaft	1
3	Bracket	2
4	Roller	1
5	Bush	1
6	Bolt(M12)	4

4. Product that is fit for purpose should have:
 - a) Accuracy of dimensions and measurements.
 - b) Correct use of symbols and conventions.
 - c) Completeness of all required details and information.
 - d) Clear labels, notes, and annotations.
 - e) Appropriate scale and proportions.
 - f) Neatness and clarity of presentation.
 - g) Compliance with relevant standards and specifications.
 - h) The drawing effectively communicates the intended design or construction requirements